

SCIENTIFIC REFEREE REPORT

The Face-Adjacency Graph of the Truncated Octahedron

Laplacian spectrum, symmetry decomposition, an inter-orbit operator, and the magnetic Laplacian

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Assessment scope: 13-page Draft 8 PDF; mathematical proofs and claims therein, version comparison with Draft 7, targeted background literature, and a fresh test of the gauge-classification hypothesis. The author’s complete tagged code release was not run.

Recommendation: Minor revision — conditional. The mathematics examined does not reveal a confirmed central error, but release-level reproducibility remains unverified and the promised permanent data/code archive must be finalized before editorial clearance.

Mathematical validity gate: No confirmed critical or major error in the examined claims. This is not a formal certification of every theorem.

Finding counts: 0 established critical; 0 established major; 2 limited corrections/closure requests; 1 material verification limitation (tagged code not executed).

Aggregate score: Intentionally omitted. The individually justified metrics and verdict, not a total, constitute this assessment.

Executive assessment

This is a strong, narrowly scoped mathematical note about an explicitly specified 14-vertex polyhedral graph. The orbit-block proof of the ordinary Laplacian spectrum, the treatment of the inter-orbit skew operator, and the exact special-flux magnetic factorizations are stated with useful and increasingly clear limits. Section 6 is the central contribution: the quarter-flux cube gauge, Gram identity, block reduction and magnetic rotation action explain spectral coincidences rather than merely listing eigenvalues.

Draft 8 differs from Draft 7 mainly by scope wording, mathematical typography, and attribution/provenance presentation; the central spectral identities are unchanged. A targeted attempt to invalidate Proposition 6.1 through its dependence on the spherical embedding did *not* find a counterexample. Exact boundary-rank analysis establishes its key cycle-space premise for the graph in question. Conversely, a torus example demonstrates why the spherical restriction must be retained.

The two remaining tasks are mainly scientific-release issues: permanent identification of the complete tagged analysis, with independently replayable results, and a more precise delimitation of the novelty claim against prior magnetic-spectral work. An incomplete release test is not evidence that the author’s programs fail. A favorable mathematical verdict must nevertheless remain conditional until that test is done.

1 Principal positive findings

1.1 Ordinary spectrum and its representation structure

The submission defines the graph on six square faces and eight hexagonal faces. Its induced hexagon graph is the cube Q_3 , while the square–hexagon incidence connects (a, σ) to the four $h \in \{\pm 1\}^3$ with $h_a = \sigma$. This is the correct starting incidence description. The decomposition

$$\mathbb{R}^{14} \cong 2A_{1g} \oplus E_g \oplus 2T_{1u} \oplus T_{2g} \oplus A_{2u}$$

is dimensionally complete. In the orthonormal T_{1u} multiplicity basis the Laplacian block is

$$\begin{pmatrix} 4 & -2 \\ -2 & 5 \end{pmatrix}, \quad \det(xI - L_{T_{1u}}) = x^2 - 9x + 16.$$

The A_{1g} block contributes $x(x-7)$; E_g, T_{2g}, A_{2u} contribute respectively $(x-4)^2, (x-7)^3, (x-9)$. Hence the manuscript's characteristic polynomial (2) follows without a 14-by-14 determinant. This algebra and the degree, dimension and trace controls are consistent with prior examined versions. It is *inherited established evidence*, not a new execution of the author's code for Draft 8.

The distinction between the T_{1u} action $K^2 = -4I$ and the A_{1g} action $K^2 = -12I$ is well drawn. Particularly valuable is the explicit warning in Remark 5.2 that K does not annihilate the full eigenvalue-7 space, since that space contains the A_{1g} component.

1.2 The magnetic spectral classes

Let A_q be the Hermitian magnetic adjacency and $\Delta_q = D - A_q$. The 24 oriented triangular plaquettes all carry phase $\exp(i\pi q/12)$. Independently of the chosen gauge, the closed-walk calculation underlying Proposition 6.2 reads

$$\begin{aligned} \text{tr}(D^3) &= 6 \cdot 4^3 + 8 \cdot 6^3 = 2112, & \text{tr}(DA_q^2) &= 6 \cdot 4^2 + 8 \cdot 6^2 = 384, \\ \text{tr}(A_q^3) &= 24 \cdot 6 \cos(\pi q/12) = 144 \cos(\pi q/12), \end{aligned}$$

and thus

$$\text{tr}(\Delta_q^3) = 3264 - 144 \cos(\pi q/12). \quad (1)$$

Equal spectra imply equal third moments, and the cosine distinguishes the representatives $q = 0, \dots, 12$. The identification $q \sim 24 - q$ follows from complex conjugation. This is a compelling and economical classification *once the spherical flux/gauge classes have been established*.

The claimed quarter-flux factorization follows from the displayed identity

$$B^*B = \frac{9I - C^2}{2} = 4(P_1 + P_2), \quad BB^* = 4I_6.$$

For cube eigenvalues $\mu = 1, -1$, the respective 2×2 blocks are

$$\begin{pmatrix} 4 & -2 \\ -2 & 5 \end{pmatrix}, \quad \begin{pmatrix} 4 & -2 \\ -2 & 7 \end{pmatrix},$$

which yield $(x^2 - 9x + 16)^3$ and $(x-3)^3(x-8)^3$, with two scalar sectors completing the count. This explains the persistent irrational pair at $q = 6$. At $q = 12$, the all-negative edge gauge gives the signless Laplacian $D + A$; the statement is exact and requires no numerical extrapolation. These results remain unchanged from Draft 7; their earlier exact corroboration is not being counted as a new clean-tag replay.

2 Targeted challenge: spherical gauge uniqueness

2.1 What Proposition 6.1 actually needs

The theorem claims that edge $U(1)$ phases on a connected graph *cellularly embedded in S^2* are determined, modulo vertex gauge, by the oriented face holonomies, subject to their product being 1. Its uniqueness direction rests on every cycle being homologous to an integer combination of oriented face boundaries. Merely counting 24 triangles is not sufficient by itself; the cycle-space spanning assertion is the critical step.

For the stated graph the following finite chain-complex calculation directly tests that step. Let C_2 have one generator for each of the 24 oriented triangles, C_1 one for each of the 36 oriented edges, and C_0 one for each of the 14 vertices. Using the actual square and hexagon coordinates of Section 2 to orient each triangle outwards, define $\partial_2 : C_2 \rightarrow C_1$ and $\partial_1 : C_1 \rightarrow C_0$. Exact integer incidence calculations give

$$\partial_1 \partial_2 = 0, \quad \text{rank}(\partial_1) = 13, \quad \text{rank}(\partial_2) = 23, \quad \dim \ker \partial_1 = 36 - 13 = 23.$$

Therefore $\text{im}(\partial_2) = \ker(\partial_1)$, and the oriented triangles span the full cycle space. Moreover the sum of the 24 outward-oriented boundaries is zero, the expected single global relation. A nontrivial vertex phase assignment was also checked to leave every triangular holonomy unchanged. The graph has vertex connectivity four, consistent with a well-defined polyhedral spherical embedding.

This *supports* the specific invocation of Proposition 6.1 and rules out the suspected cycle-space hole for the supplied graph. It is an exact finite calculation, not a machine-checked general theorem about all S^2 cellular embeddings.

2.2 Negative control and the domain boundary

The spherical hypothesis does substantive work. On a periodic 4×4 toroidal square lattice, assign a phase $e^{i\theta/4}$ along each positive horizontal edge and phase 1 on vertical edges. Every elementary face has trivial holonomy since the horizontal contributions cancel, but a noncontractible horizontal winding loop has holonomy $e^{i\theta}$. With $\theta = \pi/7$ this is nontrivial and cannot be gauged away by vertex phases. Hence the corresponding unrestricted assertion on a torus is false. The counterexample is *outside* Draft 8's stated domain and is not a manuscript error.

Verdict on the challenged claim: no defect established. The proof is valid in the specified topological setting; preserving the explicit S^2 assumption is mandatory. A useful small enhancement would be to state directly that the 24 triangular faces generate the cycle space with one relation, as the manuscript already indicates numerically.

3 Targeted prior-art assessment

The existing object is known. MathWorld's *Tetrakis Hexahedral Graph* entry describes the same 14-vertex, 36-edge graph, its automorphism group of order 48 and the same adjacency characteristic polynomial (Weisstein, reference [R1]). The submitted paper acknowledges the graph and adjacency polynomial as pre-existing, which is appropriate.

The closest inspected magnetic-spectral antecedent is Avishai and Luck (2008), reference [R2], on monopole-threaded spherical polyhedral graphs. It develops the same broad $U(1)$ link/face flux setting and reports exact spectral expressions for several classes. Its examples are the five Platonic graphs, C_{60} , N -diamonds (two N -valent poles and N degree-four equatorial vertices), and N -prisms. Those are not the degree sequence $4^6 6^8$ of the tetrakis-hexahedral graph. In the inspected source no explicit factorization equivalent to Draft 8's $q = 6$ formula or its magnetic T_1 intertwiner was located.

A focused set of searches combining *tetrakis hexahedral graph*, *magnetic flux*, *monopole*, *Laplacian spectrum*, and the factor $x^2 - 9x + 16$ produced no identified exact predecessor within this search scope. This is a **bounded non-detection**, not a claim of exhaustive discovery or priority. The paper's modest novelty language is justified. The most defensible contribution claim is the explicit orbit-sensitive Laplacian analysis and the particular exact magnetic factorizations and intertwiner for this known graph, not invention of the spherical-flux method itself.

4 Computational reproducibility and revision comparison

The attached Draft 8 PDF names tag `face-graph-note-v8`, provides a four-script execution recipe and describes certificate generation, exact integer/Gaussian-integer computations, a numerical flux sweep and previously filed independent checkers. It states that a permanent Zenodo archive will accompany submission but gives no final identifier. The figures are correctly described as ranked-eigenvalue guides, not analytic spectral branches; Section 8 distinguishes exact, numerical and human-readable proofs, and Table 2 discloses that nothing has been machine checked by Lean.

A source-level comparison of Drafts 7 and 8 finds no change to the main characteristic polynomials, the spherical-gauge theorem, the quarter-flux Gram formula, the magnetic representation or the Fedorov table. The principal visible changes remove a speculative-motivation aside, sharpen the scope statement, improve typesetting and update acknowledgements. There is therefore no scientific basis to treat inherited exact checks as newly executed tests merely because the manuscript has a higher draft number.

Release limitation: The complete immutable `face-graph-note-v8` source bundle could not be obtained in executable form in the review environment. One raw text file could be inspected remotely; neither the full tag identity nor its environment, runner exit status, regenerated certificate, CSV and figure was independently replayed. In particular, **the author release is not certified as reproducible by this review**. That is a limitation of available evidence, not a finding that its programs are faulty. The appropriate closure is a clean-environment clone/archive checkout at the exact tagged commit, dependency installation, full unoptimized runner execution, failure-injection check, and hash comparison of generated artifacts against the deposit.

5 Fourteen-metric scientific assessment

Values are diagnostic judgments on a ten-point scale, not probabilities. This specialist note is evaluated under the theoretical mathematical-physics profile (core category weights: correctness 40%, completeness 25%, clarity 20%, significance 15%). An aggregate number is not used. The within-category emphasis follows the defined review rubric.

ID	Criterion	Score	Manuscript-specific justification
M1	Theorem and claim precision	9.0	Main results are stated with well-identified graph, flux and representation conditions.
M2	Proof and support completeness	9.0	Orbit-block and Gram calculations provide intelligible proofs; tagged release execution is outstanding corroboration, not a missing theorem.
M3	Calculation and analysis accuracy	9.0	Main formulas remain consistent with exact earlier checks; new gauge incidence ranks agree. Does not imply entire release executed.
M4	Logical consistency	9.0	Spherical gauge uniqueness, magnetic charge equivalence, and third-moment distinction fit together without a detected contradiction.
M5	Definition coverage	9.0	Square/hexagon orbits, projectors, operator K , charge sector and cube spectral projectors are explicit.
M6	Assumption transparency	9.0	S^2 cellular embedding, flux orientation, proof versus computation and gauge choice are disclosed.
M7	Scope specification	8.5	Strong limits on numerical branches and physical interpretation; reproducibility claims require a checked immutable release.
M8	Notation consistency	8.5	Representation labels and q -sectors are mostly stable; a few dense block/figure expressions may slow first reading.

ID	Criterion	Score	Manuscript-specific justification
M9	Proof accessibility	9.0	Explicit small matrices and constructive gauges make sophisticated ideas auditable by a trained reader.
M10	Organization/navigation	8.5	Progression from graph to spectra to flux and Fedorov table is coherent; presentation improvements are visible.
M11	Originality	7.0 (prov.)	Distinguishable exact magnetic examples not located in a bounded search; known graph and known general monopole framework clearly separated.
M12	Generality	6.5	Results primarily concern one graph and two exactly solved fluxes, though methods suggest transfer.
M13	Literature/attribution	8.0	Prior graph and monopole work acknowledged; targeted comparison can be made more explicit in the article.
M14	Potential impact	7.0	Valuable specialist contribution in spectral graph theory and magnetic polyhedral models; broad physical applications are not established here.

Category interpretation: mathematical correctness and supporting proofs are strong; completeness is high but external computational corroboration remains conditional; clarity is strong with minor presentational demands; significance is that of a potentially original *specialist example*, not a demonstrated general principle.

Specialized checks (not added to an aggregate): M15 Gauge/invariant structure 9.0/10 — the crucial spherical topology premise passed the targeted challenge; M18 Algebraic structure 9.0/10 — the orbit blocks and quarter-flux representation are explicit. M16 (claimed breakthrough) and M17 (interdisciplinary synthesis) are not activated by this deliberately mathematical note.

6 Prioritized corrections and dependency effects

C1 — Release-level reproduction [closure required; not a proven code defect]. Section 8, pages 10–11: deposit the exact release/tag/commit with pinned dependency metadata, figure data and regenerable certificates; verify its one-command runner in a clean environment and publish its immutable archive identifier. A failure to regenerate the $q = 6$ and $q = 12$ polynomials or correct holonomies would affect Theorem 6.4’s computational corroboration; it would not by itself refute the written hand proof. A failure revealing mismatch between the defined graph/gauge and the manuscript equations would be more consequential and require rereview.

C2 — Precision of priority and context [minor scholarly revision]. Section 1, pages 1–2, and references: add a sentence explicitly distinguishing the previously catalogued tetrakis-hexahedral adjacency graph and Avishai–Luck monopole methods from the present graph-specific Laplacian and quarter-flux calculations. No new priority claim is needed. An exact previously published matching factorization would require revising originality and contribution wording, but would not automatically invalidate the mathematics.

Optional clarification — the topology premise. Proposition 6.1, pages 5–6: keep the restriction to a *cellular embedding on S^2* prominent; the triangle rank-23 explanation could be promoted into one additional sentence. The present proof and the negative control support, rather than undermine, the statement as written.

The ordinary spectrum, K decomposition, face-support weights, and Fedorov table are not logically dependent on the magnetic gauge-classification step and remain independently interpretable even if a future magnetic implementation discrepancy is found. Proposition 6.2’s count of 13 distinct spectra

depends on the magnetic gauge-class premise and equation (4); the special-flux theorems depend on the explicit phase construction and block factorizations. These links determine the recheck priority after any change.

7 Recommendation to authors and editor

To the authors. Complete C1 first. Identify the exact source tree, pinned runtime, and outputs; submit a clean verification receipt alongside the immutable deposit. Make the small contextual amendment C2. Avoid converting the numerical connecting segments of Figure 1 into physical band branches, and retain the separation between finite exact computation and formal proof.

To the editor. Conditional minor revision is justified on the mathematics inspected and the currently narrow claims. Do not treat the tag replay as successful. Final clearance should require the archived release to exist and the declared runner to succeed or a disclosed explanation of any mismatch. The novelty judgment is favorable but deliberately bounded; venue fit and final priority assessment remain editorial questions.

What would reverse this recommendation? A validated counterexample to Proposition 6.1 *within the stated spherical cellular-embedding domain*, or a reproducible disagreement between the graph/gauge specified in equations (7)–(9) and the claimed characteristic polynomial in (5), would change this to major revision or rejection of the present proof dependent on the defect. A reproducibility failure that the authors cannot remedy would preclude final acceptance pending repair. Discovery of identical earlier magnetic results would alter the originality/venue judgment, not necessarily mathematical validity. Conversely, a successful immutable clean-tag replay and C2 closure would remove the remaining verification reservation.

8 Final judgment

Draft 8 presents coherent and well-motivated finite mathematics on the tetrakis-hexahedral face graph. A fresh, adversarial test strengthens confidence in the most consequential topological premise, and the targeted literature comparison supports a specific, carefully qualified possible contribution. The strongest objection left is evidentiary rather than algebraic: the full published software release has not been independently executed. **Recommendation: minor revision conditional on reproducibility closure; mathematical validity not blocked within the examined scope.**

References inspected for this review

- [R1] E. W. Weisstein, *Tetrakis Hexahedral Graph*, Wolfram MathWorld, <https://mathworld.wolfram.com/TetrakisHexahedralGraph.html>. Catalogued graph, automorphism order and ordinary adjacency polynomial.
- [R2] Y. Avishai and J. M. Luck, *Tight-binding electronic spectra on graphs with spherical topology I: The effect of a magnetic charge*, J. Stat. Mech. (2008) P06007, DOI [10.1088/1742-5468/2008/06/P06007](https://doi.org/10.1088/1742-5468/2008/06/P06007). Closely related magnetic-flux setting and examined examples.